Banking Analytics Case Study

Project Overview

This case study presents a **Banking Data Analysis** project developed using **SQL, Python, Excel, and Power BI**. The project analyzes banking operations using **4 interconnected datasets**: Customers, Accounts, Loans, and Transactions. The datasets include **5,000 customer records, 5,000 account records, 5,000 loan records, and 100,000 transaction records**, enabling large-scale business analysis.

The project follows an end-to-end data analytics workflow, including data cleaning, SQL analysis, exploratory data analysis (EDA), statistical analysis, customer segmentation, and interactive dashboard development.

Analysis Objective

The objective of this project was to analyze banking data to:

- Understand customer distribution across different cities, occupations, and account types.
- Analyze account balances and identify customer segments based on banking activity.
- Examine loan distribution across different loan types, interest rates, and loan status.
- Study transaction patterns across various banking channels, months, and customer locations.
- Develop interactive dashboards and KPI reports to support data-driven decision-making.

Business Problem

The bank wanted to convert raw customer, account, loan, and transaction data into meaningful insights. The goal was to build an interactive dashboard that helps managers monitor business performance and make data-driven decisions.

Dataset Description

- **Total Customers:** 5,000
- **Total Accounts:** 5,000
- **Total Loans:** 5,000
- **Total Transactions:** 100,000

Key Attributes

- Customer_ID
- Customer_Name
- Account_ID
- Account_Type
- Balance
- Loan_ID
- Loan_Type
- Loan_Amount
- Interest_Rate
- Transaction_ID
- Transaction_Date
- Transaction_Channel
- Transaction_Amount
- City

The dataset consists of four interconnected tables linked through primary and foreign keys, enabling comprehensive analysis of customer behavior, account performance, loan portfolios, and transaction activities. Before analysis, the data was cleaned, validated, and prepared to ensure consistency across all datasets.

Data Dictionary

Column	Description
Customer_ID	Unique identifier for each customer
Account_ID	Unique identifier for each bank account
Account_Type	Type of account (Savings, Salary, Current)
Balance	Current account balance
Credit_Score	Customer's creditworthiness score
Loan_Amount	Amount of loan sanctioned to the customer
Loan_Status	Current status of the loan (Active/Closed/Default)
Transaction_Amount	Amount involved in each banking transaction
Transaction_Type	Type of transaction (UPI, NEFT, IMPS, etc.)
Branch_ID	Unique identifier of the bank branch where the account is maintained

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Dataset Understanding

Table	Purpose
Customers	Customer demographic and banking details
Accounts	Account information and balances
Transactions	Deposits, withdrawals, transfers
Loans	Loan amount, type, repayment status

Tools & Technologies Used

- **Programming Language:** Python
- **Libraries:**
  - **Pandas** – Data cleaning and analysis
  - **NumPy** – Numerical operations
  - **Matplotlib & Seaborn** – Data visualization
- **Database:** MySQL – Data storage and SQL analysis
- **SQL:** Joins, Aggregate Functions, Window Functions, CTEs
- **Data Handling:** CSV Files
- **Google Sheets:** Formulas, Pivot Tables, Charts
- **Power BI:** Interactive dashboards and KPI reporting

Data Preparation & Preprocessing

The banking datasets were prepared before analysis to ensure accurate and reliable results.

- Removed duplicate records from the datasets.
- Checked and handled missing or invalid values.
- Corrected data types for dates, numbers, and text fields.
- Standardized column names and formatting across all datasets.
- Validated relationships between Customers, Accounts, Loans, and Transactions using common keys.
- Imported the cleaned datasets into MySQL, Excel, Python, and Power BI for analysis.

Analysis Performed

The banking dataset was analyzed using SQL, Python, Excel, and Power BI to identify trends, measure business performance, and generate actionable insights.

1. Customer Analysis

- Analyzed customer distribution by city, occupation, and gender.
- Identified cities with the highest customer base.

2. Account Analysis

- Analyzed account types and account balances.
- Identified customers with the highest account balances.

3. Loan Analysis

- Compared loan amounts across different loan types.
- Analyzed loan status and interest rate distribution.

4. Transaction Analysis

- Analyzed transaction volume by channel and month.
- Identified the most frequently used banking channels and monthly transaction trends.

5. Dashboard & KPI Analysis

- Built KPIs and interactive dashboards using Excel and Power BI.
- Visualized key business metrics to support decision-making.

Key Observations & Business Relevance

Key Observations

- Savings accounts contributed the highest overall account balance among all account types.
- Mobile banking was the most frequently used transaction channel, reflecting strong digital banking adoption.
- Home loans represented the largest share of the bank's loan portfolio.
- Customer distribution varied across cities, with a few cities contributing a larger customer base.
- Monthly transaction analysis highlighted fluctuations in transaction values throughout the year.

Business Relevance

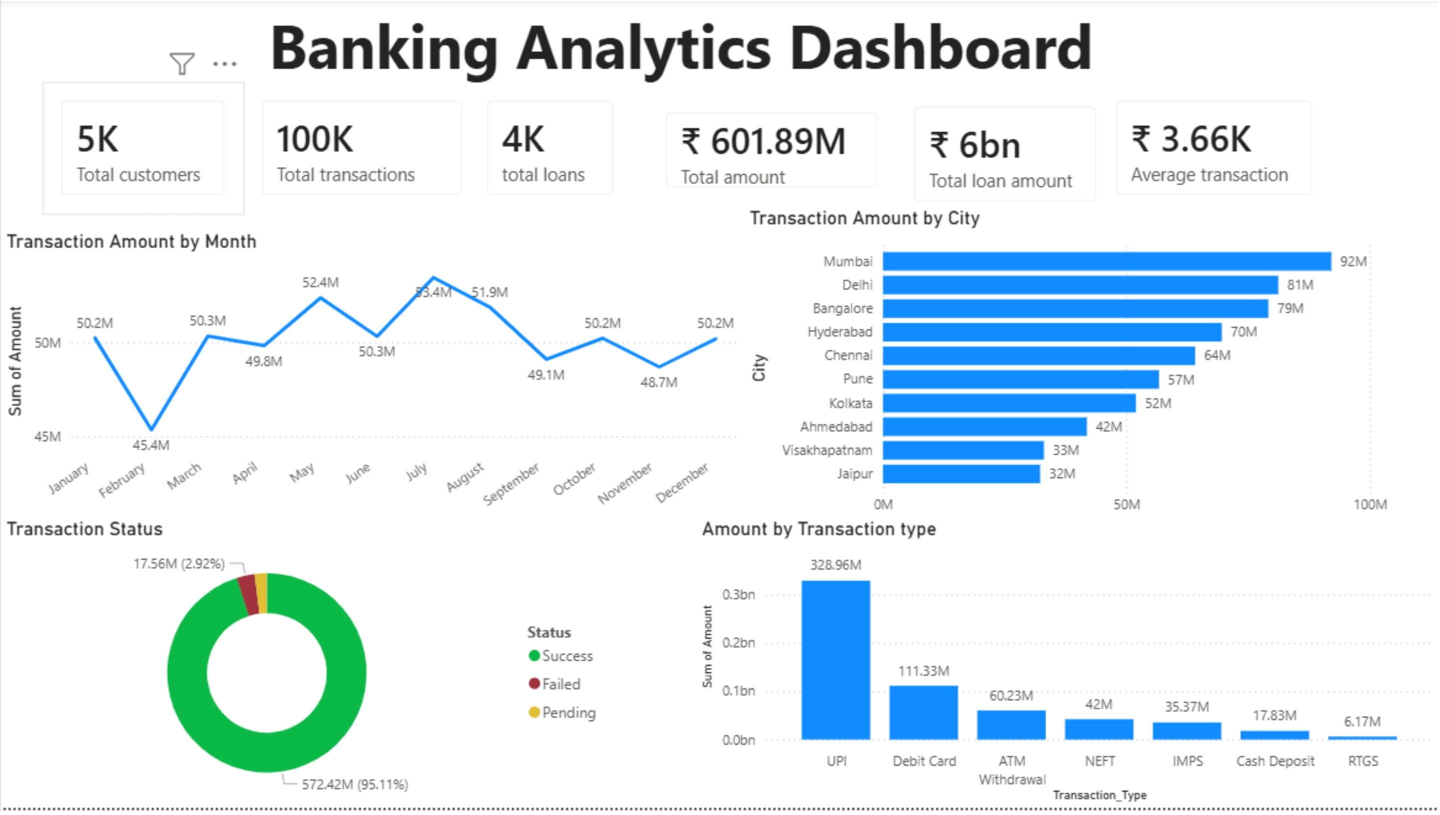
The analysis provides valuable insights that can support banking operations and business decision-making by:

- Identifying high-value customers for premium banking services.
- Understanding customer preferences across different banking channels.
- Monitoring loan portfolio performance and customer borrowing trends.
- Supporting management with KPI-driven reports and interactive dashboards.

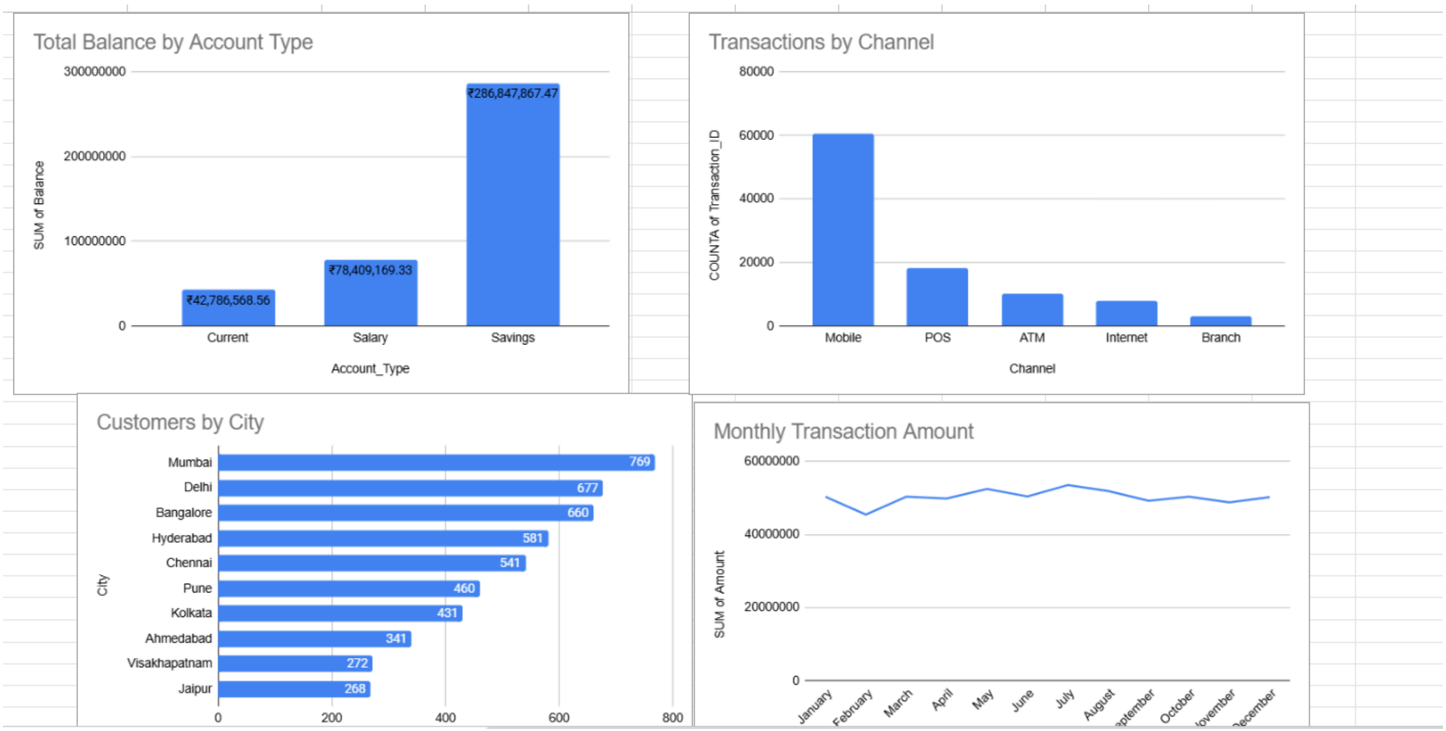
Banking Data Analysis Project | SQL, Python, Excel, Power BI

- Performed end-to-end analysis on **115,000+ banking records** across **Customers, Accounts, Loans, and Transactions** datasets.
- Cleaned, validated, and prepared banking data using **SQL, Python (Pandas), and Power Query** to ensure data quality and consistency.
- Developed **SQL business solutions** using **Joins, CTEs, Window Functions, CASE statements, Aggregate Functions, Running Totals, and Branch Performance Analysis**.
- Conducted **Exploratory Data Analysis (EDA)** and **Customer Segmentation** to identify customer behavior, account trends, and high-value customer groups.
- Performed **Correlation Analysis** in Excel to evaluate relationships between key banking metrics as part of statistical analysis.
- Built interactive **Power BI dashboards** using **Data Modeling, DAX, KPI Cards, Slicers, and multiple report pages** to monitor branch performance, customer activity, loans, and transactions.
- Generated business insights and recommendations to support data-driven decision-making in customer management, branch performance, and banking operations.

Power BI:


Banking Analytics Dashboard
5K
Total customers
100K
Total transactions
4K
total loans
₹ 601.89M
Total amount
₹ 6bn
Total loan amount
₹ 3.66K
Average transaction
Transaction Amount by Month
Sum of Amount
50M
45M
50.2M
45.4M
50.3M
49.8M
52.4M
50.3M
53.4M
51.9M
49.1M
50.2M
48.7M
50.2M
January
February
March
April
May
June
July
August
September
October
November
December
Transaction Amount by City
City
Mumbai 92M
Delhi 81M
Bangalore 79M
Hyderabad 70M
Chennai 64M
Pune 57M
Kolkata 52M
Ahmedabad 42M
Visakhapatnam 33M
Jaipur 32M
0M
50M
100M
Transaction Status
17.56M (2.92%)
572.42M (95.11%)
Status
Success
Failed
Pending
Amount by Transaction type
Sum of Amount
0.3bn
0.2bn
0.1bn
0.0bn
UPI 328.96M
Debit Card 111.33M
ATM Withdrawal 60.23M
NEFT 42M
IMPS 35.37M
Cash Deposit 17.83M
RTGS 6.17M
Transaction_Type

Sheets:


Total Balance by Account Type
SUM of Balance
300000000
200000000
100000000
0
Current ₹42,786,568.56
Salary ₹78,409,169.33
Savings ₹286,847,867.47
Account_Type
Transactions by Channel
COUNTA of Transaction_ID
80000
60000
40000
20000
0
Mobile
POS
ATM
Internet
Branch
Channel
Customers by City
City
Mumbai 769
Delhi 677
Bangalore 660
Hyderabad 581
Chennai 541
Pune 460
Kolkata 431
Ahmedabad 341
Visakhapatnam 272
Jaipur 268
0
200
400
600
800
Monthly Transaction Amount
SUM of Amount
60000000
40000000
20000000
0
January
February
March
April
May
June
July
August
October

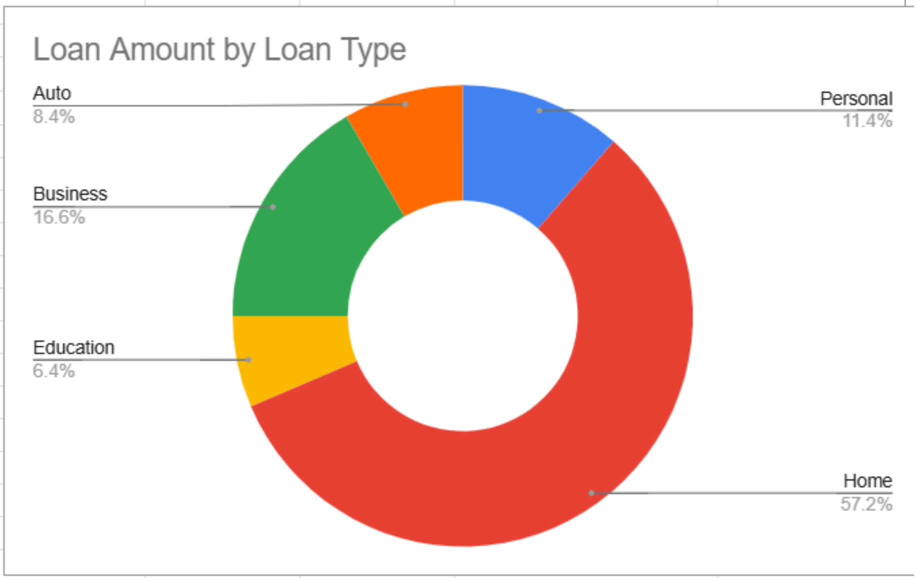
Account_Type	SUM of Balance
Current	42786568.56
Salary	78409169.33
Savings	286847867.5

City	COUNTA of Customer_ID
Mumbai	769
Delhi	677
Bangalore	660
Hyderabad	581
Chennai	541
Pune	460
Kolkata	431
Ahmedabad	341
Visakhapatnam	272
Jaipur	268

Loan_Type	SUM of Loan_Amount
Personal	645934439
Home	3246930247
Education	363552024
Business	940108970
Auto	479404307

Channel	COUNTA of Transaction_ID
Mobile	60729
POS	18178
ATM	10141
Internet	7991
Branch	2961

Month number	Month	SUM of Amount
1	January	50337248.83
2	February	45452186.07
3	March	50378648.09
4	April	49868849.55
5	May	52459446.67
6	June	50425614.11
7	July	53566252.41
8	August	51939814.23
9	September	49248537.84
10	October	50381029.66
11	November	48790701.16
12	December	50273268.45



Conclusion

This **Banking Data Analysis** project provided hands-on experience in analyzing **115,000+ banking records** across customer, account, loan, and transaction datasets using **SQL, Python, Excel, and Power BI**. The project involved data cleaning, validation, exploratory data analysis

(EDA), customer segmentation, statistical analysis, and dashboard development to transform raw banking data into meaningful business insights.

Using **advanced SQL techniques**, **Python for data analysis**, **Excel for reporting and correlation analysis**, and **Power BI for interactive dashboards**, the project delivered insights into customer behavior, branch performance, account balances, loan portfolios, and transaction trends. The analysis also supported business recommendations to improve decision-making and operational performance.

Overall, this project strengthened my understanding of the complete data analytics lifecycle—from data preparation and analysis to visualization and business storytelling—and demonstrated my ability to solve real-world business problems using data-driven approaches.